

Phase 1–3 Implementation Gap Analysis Report

Client Requirements vs. Current Codebase — Evidence-Based Audit

Project	OpenClaw — AI Content Platform (OGz Studios, Riyadh)
Repository	OGz-Studios monorepo (branch: faisal-dev8)
Requirement sources	OpenClaw_TechDoc_v1.0 (43 pp) · OGZ_Simple "What We Built" (18 pp)
Scope of audit	Phase 1, Phase 2, Phase 3
Report date	13 June 2026
Method	Full read of both docs + direct inspection of source, migrations, n8n flows

1. Executive Summary

This report compares the two client requirement documents against the actual state of the OpenClaw codebase. The **OpenClaw_TechDoc_v1.0** is treated as the authoritative contractual baseline — it explicitly states it is the *"single source of truth for OpenClaw Phase 1 architecture"* (§14.3) and defines the Phase Roadmap (§1.3). The **OGZ_Simple** document is a later "What We Built" narrative (June 2026, v2.0) describing an *evolved* creative direction (5 Creative-Director brains, a 3-model cross-compare, 94 visual chains, fal.ai); it is used here as a secondary reference and a source of intent for several Phase 2/3 items.



Headline findings:

- **Phase 1 is functionally delivered.** All 31 BrandDNA / system tables, RLS & append-only policies, PDPL cascade, all 7 client screens, all 9 admin views, 3 Copilots, the full CEO → COO → DeepSeek → CCO → CEO AI chain, the 11 human-override triggers, the Memory Controller, and all 11 canonical n8n flows exist with evidence.
- **Two large architectural deviations from the TechDoc** (both deliberate, documented in code): (1) the visual layer uses **fal.ai**, not the spec's **Weavy.ai**; (2) the CEO/COO agents run on **OpenAI / DeepSeek**, not the spec's **Claude Sonnet 4.6 / Haiku 4.5** (Anthropic budget). Functionality is preserved; the contract text is not.
- **Genuine Phase 1 gaps:** Stripe payments (stub returning HTTP 501), PostHog analytics (empty package), and a hard-coded-secrets problem in the n8n flows that violates security requirement SEC-02.
- **The OGZ_Simple "5 Creative Director brains / cross-compare / ALLaM 34B" system is not implemented** as described; the code instead uses a deterministic `composition_matrix` of 6 creative methods plus a visual-chain library.

Status legend: COMPLETE built & evidenced PARTIAL present but incomplete MISSING not implemented DEVIATION built differently than spec N/A NOW not expected in this phase

2. Methodology & Evidence Basis

Conclusions are drawn only from (a) the two requirement PDFs and (b) direct inspection of the repository. The audit covered:

- **Database:** `supabase/migrations/*` (111 migrations), `packages/db`, `sql/`.
- **Frontend:** `apps/web/src/app` (75 page routes, 70+ API routes), layouts, i18n.
- **AI agents:** `packages/ai/src/providers`, `packages/memory`, `prompts/`.
- **Orchestration:** `n8n/flows` (17 flow exports), `packages/n8n`.
- **Integrations & services:** `packages/{image,scraping,payments,email,vectors,analytics,compliance,pdpl}`, `services/{cio-analyzer,generation-worker,pdpl-janitor}`.
- **Project record:** `CHANGELOG.md`, which documents the deliberate architectural decisions cited below.

Note on the two documents. The TechDoc (April 2026) specifies Weavy.ai, Claude-based C-suite, DeepSeek captions and a single-caption pipeline. The OGZ_Simple deck (June 2026) describes a re-architected creative engine (fal.ai, 5 CD brains, 3-model comparison, 94 chains). The codebase sits *between* the two: it kept the TechDoc's governance/data architecture but adopted the OGZ_Simple visual stack (fal.ai + chain library) and a budget-driven model swap. Several "gaps" below are therefore really *deviations* where the implementation followed the newer direction rather than the older contract.

3. Phase 1 — Detailed Findings

TechDoc §1.3 Phase 1 scope: *Onboarding, calendar generation, download delivery, admin panel, 3 Copilots, Arabic QC. Agents: CEO, COO, CCO, DeepSeek.* Plus the full Phase-1 tech stack (§2.1), 7-layer architecture (§3), BrandDNA schema (§4), 11 n8n flows (§5), AI integration specs (§6), DB/RLS (§7), 7 client screens + admin (§8), security checklist SEC-01...10 (§9).

3.1 Database & BrandDNA Schema — COMPLETE

Requirement (TechDoc §4, §7)	Status	Evidence / Notes
Layer 1 private-brand tables (11)	COMPLETE	brand_profiles, audience_profiles, visual_style_profiles, channel_profiles, evidence_bundles, source_records, negative_patterns, override_rules, onboarding_responses, brand_performance_log, brand_snapshots — <code>supabase/migrations/0001_init.sql</code>
Layer 2 sector intelligence (3)	COMPLETE	sector_baselines, sector_question_weights, sector_trends — <code>0001_init.sql:97–121</code>
Layer 3 global intelligence	COMPLETE	content_performance_patterns, onboarding_intelligence, visual_performance_global, occasion_intelligence. <code>negative_pattern_library</code> replaced by <code>global_negative_patterns</code> (0041/0042) — equivalent.
Append-only audit tables	COMPLETE	routing_decisions & branddna_event_log with <code>no_update / no_delete</code> RLS policies — <code>0001_init.sql:579–600</code>
Memory Controller queue + governance tables	COMPLETE	memory_controller_queue, confidence_classifications, qa_review_queue, usage_logs, anomaly_records, calendars/calendar_posts, deletion_audit_log
RLS on ALL tables + client isolation	COMPLETE	Per-table policies; client read via <code>brand_id = auth.uid()</code> join — <code>0001_init.sql:492–625</code>
PDPL two-phase cascade delete + audit log	COMPLETE	<code>cascade_delete_brand()</code> anonymises append-only tables, deletes Layer 1, logs Phase 1 — <code>0005_pdpl_cascade.sql</code> . (Phase-2 external cleanup janitor is stubbed — see 3.6.)
Critical indexes	COMPLETE	11 indexes incl. brand_id, shard, qa-queue, event-log — <code>0001_init.sql:475–486</code>

The schema has grown well beyond the spec (22+ extra tables for chains, competitors, copilot threads, method profiles, occasion events, on-demand requests) — none of the spec tables are missing.

3.2 Frontend — Client Platform & Admin — COMPLETE (2 partials)

Screen / View (TechDoc §8.4)	Status	Evidence / Notes
1. Sign Up (email + Google OAuth, RTL, slug)	COMPLETE	<code>app/[slug]/auth/page.tsx</code> , <code>{auth}/*</code> , <code>/api/auth/callback</code>
2. Onboarding 15-question form	PARTIAL	Original single-page form replaced by a multi-stage <i>onboarding-start</i> → <i>extraction</i> → <i>review</i> flow; <code>app/[slug]/onboarding/page.tsx</code> now redirects. Functionally present but the "15 questions all-at-once" UX described in §8.4 was re-architected.
3. Processing (realtime, 10-min timeout)	COMPLETE	Supabase realtime subscription to brand_snapshots — <code>app/[slug]/processing/page.tsx</code>
4. Brand Snapshot	COMPLETE	Voice, visual style, audience, completeness, dialect, occasions — <code>app/[slug]/snapshot/page.tsx</code>
5. Calendar Dashboard (8 free, 9–20 blurred, download, watermark badge)	COMPLETE	<code>app/[slug]/calendar/[month]/page.tsx</code> + 3-month rolling view
6. Brand Profile (completeness, correction form)	COMPLETE	Field-by-field confidence badges, correction form, event log — <code>app/[slug]/profile/page.tsx</code>
7. Upgrade Flow (Stripe Checkout)	PARTIAL	Plan cards render (Free / Starter / Pro) but Stripe Checkout is not wired; webhook returns 501 — see 3.6. UI-only.
Admin panel — 9 views	COMPLETE	Client list, QA queue, Cost monitor, Anomalies, Routing log, BrandDNA inspector, Flow status, Sector baselines, Occasion calendar — all under <code>app/admin/*</code> (plus ~20 extra admin views)

3 Admin Copilots (management / tech / production)	COMPLETE	SSE streaming, RLS-scoped context, rate-limited — <code>app/admin/copilot/*</code> , <code>/api/copilot/[role]/route.ts</code>
Arabic RTL by default + locale toggle	COMPLETE	<code><html dir=rtl lang=ar></code> in <code>app/layout.tsx</code> ; cookie-persisted toggle; Arabic fonts loaded

3.3 AI C-Suite & Governance — COMPLETE · MODEL DEVIATION

Agent	Spec model (§2.1)	Actual model	Status & evidence
CEO — routing, gating, governance	Claude Sonnet 4.6	OpenAI GPT-5 / 4o / 5-nano	DEVIATION Logic complete (classify, confidence gate, dispatch, nominations) — <code>packages/ai/src/providers/ceo.ts</code> . Comment: "Migrated from Claude ... Anthropic over budget."
COO — BrandDNA, CaptionContext, scoring	Claude Haiku 4.5	DeepSeek chat / OpenAI fallback	DEVIATION All 4 jobs implemented — <code>coo.ts</code> . Same budget migration note.
CCO — Arabic QC, flags, 0–100 score	GPT-5 (4o fallback)	OpenAI GPT-5 / 4o	COMPLETE Matches spec — <code>cco.ts</code>
DeepSeek — 20 Arabic captions	DeepSeek V3	deepseek-chat	COMPLETE — <code>deepseek.ts</code>
Memory Controller — sole BrandDNA writer	—	—	COMPLETE Queue drain, field whitelist, PII scrubber, idempotency, event-log audit — <code>packages/memory/src/{index,process}.ts</code>
11 Human-Override Triggers + Confidence Gate	—	—	COMPLETE All 11 coded as deterministic post-LLM checks — <code>packages/ai/src/human-overrides.ts</code>
3 Copilots (Sonnet 4.6)	Claude Sonnet 4.6	claude-sonnet-4-6	COMPLETE — <code>copilot.ts</code>

Hard-rule compliance (TechDoc §1.4): Memory Controller as sole writer ✓; Arabic never in image prompts ✓ (regex guard throws in `packages/image/src/visual-prompt.ts`); CDN URLs never stored ✓ (download → Supabase in `storage.ts`); only `@repo/ai/prompts` reads `*_SYSTEM_PROMPT` ✓. **The governance contract is intact** — only the underlying model vendors changed.

3.4 n8n Orchestration — COMPLETE · PRODUCTION-READINESS GAPS

Flow	Status	Evidence / Notes (all in <code>n8n/flows/</code>)
A01 batch calendar	COMPLETE	Full chain + V01 sub-flow; retry/backoff, usage_logs, S03 error route
A02 on-demand single post	PARTIAL	Flow exists but filename/name/webhook path say "Testing Mode" (<code>/webhook-test/openclaw-a02</code>) — must be promoted to production
A03 onboarding + 3 extractors	COMPLETE	Apify IG + website + Google Places
A04 brand correction	COMPLETE	CEO classify → Memory Controller nomination
A05 monthly upgrade readiness	COMPLETE	COO upgrade scoring
B03 revision request	COMPLETE	CEO → COO → DeepSeek → CCO → CEO
V01 visual chain + Sharp overlay	DEVIATION	Implemented via <code>fal.ai</code> (<code>fal-ai/flux-pro/v1.1-ultra</code>), not Weavy.ai. Sharp Arabic overlay + watermark present. + parallel <code>V01-Worker</code> .
D02 maintenance + CIO trigger	COMPLETE	1st-of-month maintenance; CIO call is Phase-2 (correctly stubbed)

S01 pipeline health (15 min)	COMPLETE	→ Tech Copilot
S02 cost ceiling 70/90/100%	COMPLETE	→ Management Copilot
S03 anomaly router	COMPLETE	Routes anomalies to correct Copilot
Extra flows (A06 extraction, A07 competitor, P01 IG publisher, QA-Approved, A02 error handler)	EXTRA	Beyond the 11; A07/P01 support Phase 2/3 work

Security gap — SEC-02 violation. Multiple n8n flow exports contain **hard-coded secrets** (OpenAI keys, Supabase service keys, webhook secrets) and temporary **ngrok URLs** inside code nodes. TechDoc §9 SEC-02 requires "Zero API keys in source code". `packages/n8n` is an empty placeholder. These must be moved to n8n credentials/env and the tunnels replaced with production URLs before launch.

3.5 Integrations — Phase 1 Stack (TechDoc §2.1)

Layer	Spec	Status	Evidence
Visual generation	Weavy.ai REST API	DEVIATION	<code>fal.ai</code> used exclusively — <code>packages/image/src/fal-client.ts</code> , <code>generate.ts</code> . No Weavy client exists. Model routing Nano-Banana/Flux preserved conceptually.
Image processing	Sharp overlay/watermark/safe-zone	COMPLETE	Dialect-aware Arabic fonts, "Beta AI Draft" watermark — <code>packages/image/src/overlay.ts</code>
Storage discipline	Supabase only, no CDN URLs in DB	COMPLETE	<code>packages/image/src/storage.ts</code> — path <code>clients/{brand_id}/calendars/...</code>
Scraping	Apify IG + website + Google Places	COMPLETE	<code>packages/scraping/src/{instagram,website,places}.ts</code>
Email	Resend + Arabic templates	COMPLETE	DB-driven templates, RTL, retry — <code>packages/email/src/notify.ts</code>
Vector DB	Qdrant per-brand namespaces	COMPLETE	<code>packages/vectors/src/index.ts</code> — <code>brand_{id}</code> collections
Payments	Stripe Checkout, SAR	MISSING	Stub. <code>packages/payments</code> empty; webhook returns HTTP 501 (Sprint S6.05) — <code>api/webhooks/stripe/route.ts</code>
Analytics	PostHog funnel tracking	MISSING	Empty package — <code>packages/analytics/src/index.ts</code> = "re-export here when modules exist"
Compliance gate	(supports Arabic QC)	COMPLETE	Deterministic pre-gen gate, religious/cultural blocks — <code>packages/compliance/src/</code>

3.6 Phase 1 — Confirmed Gaps & Incomplete Items

Item	Status	Gap description & impact
Stripe payments / Upgrade flow	MISSING	No Checkout session creation, no webhook signature verification, no tier-update on payment. The whole monetisation path (Screen 7 + tier unlock for posts 9–20) is non-functional. Blocks revenue and SEC-10 (Stripe webhook verification) cannot be tested.
PostHog analytics	MISSING	No event tracking / funnel / upgrade triggers. TechDoc lists PostHog in the Phase 1 stack. No instrumentation exists.
Hard-coded secrets in n8n (SEC-02)	RISK	Live API keys + ngrok tunnels embedded in flow JSON. Security non-compliance; tunnels will break in production.

A02 on-demand flow in Testing Mode	PARTIAL	Not on a production webhook path / name.
Onboarding form UX re-architected	PARTIAL	Delivered as a multi-stage flow rather than the single 15-question page in §8.4. Verify this matches client acceptance criteria.
PDPL external-cleanup janitor	PARTIAL	SQL Phase-1 cascade exists; the Phase-2 async Qdrant/Storage cleanup worker (<code>services/pdpl-janitor</code>) throws "implement in sprint S8.08". SEC-05 only partially closed.
Model vendor vs. contract (CEO/COO)	DEVIATION	Running on OpenAI/DeepSeek instead of Claude Sonnet/Haiku. Needs client sign-off or revert; affects §13.1 acceptance and prompt-caching economics.
Weavy vs. fal.ai	DEVIATION	Entire visual stack switched providers. Update the contract/ADR (CHANGELOG) so milestone acceptance reflects fal.ai.

4. Phase 2 — Detailed Findings

TechDoc §1.3 Phase 2 scope: *Postiz auto-publishing, Meta Insights API, CIO (Gemini 2.5 Pro) pattern analysis, client OAuth*. Status in spec: "Architecture Ready" (not built in Phase 1).

Requirement	Status	Evidence / Notes
Postiz auto-publishing	PARTIAL	OAuth connect, integration detect, channel sync and a signed webhook all exist — <code>api/postiz/{connect,detect,sync-channels,webhook}/route.ts</code> — and an <code>N8N-P01-instagram-publisher</code> flow exists. But the approved-post → scheduled-publish pipeline is not yet wired end-to-end. Connection layer done; publishing automation incomplete.
CIO — Gemini 2.5 Pro pattern analysis	MISSING	<code>services/cio-analyzer/src/index.ts</code> throws "Phase 2 placeholder". No Gemini integration. D02 flow has the trigger hook ready. (Expected per phasing, but not started.)
Meta Insights API	MISSING	Zero references to Meta/Instagram Insights or Graph API in the codebase.
Client Google Business OAuth	MISSING	Only the public Google Places lookup exists (Phase 1 onboarding). No Business-Profile OAuth or full GBP data pull. A schema slot (<code>google_business_extraction</code>) is reserved but unused.
brand_performance_log ingestion (for Meta data)	PARTIAL	Table exists (Phase-1 foundation), but no performance-data ingestion flow is wired.

Phase 2 verdict (~30%). The *foundations* the TechDoc promised (calendar platform fields, Layer 2/3 tables, D02 hook, channel_profiles) are in place, and Postiz connectivity is genuinely started. The three analytical/integration deliverables (CIO, Meta Insights, Business OAuth) are not implemented.

5. Phase 3 — Detailed Findings

TechDoc §1.3 Phase 3 scope: *CARO Arabic QC (Jais/DeepSeek), video/motion chains, advanced competitor analysis*. New agent: CARO. Status in spec: "Planned".

Requirement	Status	Evidence / Notes
Video / motion generation chains	COMPLETE	Built ahead of schedule. Two-model fal.ai pipeline (Flux keyframe → Kling/Seedance animate), async submission + webhook callback, fallback to static image, <code>calendar_posts.video_status</code> tracking — <code>packages/image/src/fal-client.ts</code> , <code>api/image/generate-video</code> , <code>api/webhooks/fal-ai/video-callback</code> . Visual-chain library (<code>chains</code> table, admin chain manager, ~88+ chains) aligns with OGZ_Simple's "94 visual chains".
CARO Arabic QC (Jais / DeepSeek)	MISSING	No CARO agent; no Jais model. Arabic QC is still performed by the GPT-based CCO. (Spec notes CCO is model-agnostic and CARO is a swap — the swap has not been done.)
Advanced competitor analysis	PARTIAL	Basic competitor capture exists — <code>N8N-A07-competitor-extraction</code> flow, <code>competitor_accounts/snapshots/alerts</code> tables, <code>admin/competitors</code> & <code>[slug]/competitors</code> pages, scorecard ranking. The <i>advanced</i> analysis engine (positioning strategy, content-gap, sentiment shift) is not implemented.
Generation worker (Phase-3 scale service)	N/A NOW	<code>services/generation-worker</code> is a placeholder (activates at >2000 brands per §11.2). Correctly deferred.

Phase 3 verdict (~40%). The most technically demanding Phase-3 item — video/motion generation — is effectively done. The two remaining items (CARO Arabic QC and deep competitor analysis) are missing or only superficially present.

6. Cross-Cutting Discrepancies (Both Documents)

6.1 The "5 Creative Director brains" engine (OGZ_Simple) is not built as described

The OGZ_Simple deck centres on a creative engine that, for every brief, generates five captions from five Saudi creative philosophies — **CD-01 Firaasa**, **CD-02 Metaphor**, **CD-03 Authenticity**, **CD-04 Heritage**, **CD-05 Paradox** — then a router scores and picks the best, and a **3-model cross-compare** (HUMAIN/ALLaM 34B vs Claude Sonnet vs GPT-4o) at `/cross-compare` feeds a learning loop toward an OGZ fine-tuned model.

OGZ_Simple claim	In code?	What actually exists
5 distinct CD brains (Firaasa...Paradox)	DIFFERENT	A deterministic <code>composition_matrix</code> of 6 creative methods (Authenticity, Vulnerability, Heritage, Paradox, Diagnostic, Metaphor) selected via COO tool calls — not 5 independent generative "brains". No "Firaasa" routing.
3-model cross-compare + <code>/cross-compare</code> UI	MISSING	No 3-model side-by-side, no <code>/cross-compare</code> page, no human-preference training-pair capture.
ALLaM / HUMAIN 34B Saudi model	MISSING	No ALLaM/HUMAIN integration anywhere.
94 visual chains (88 + 6), image+video, fal.ai	PARTIAL	Visual-chain library + fal.ai exist (<code>chains</code> table, admin manager, video pipeline). The full 94-chain / 23-family catalogue should be verified for completeness against the deck.
"1,000 clicks → OGZ's own fine-tuned Saudi AI"	MISSING	No preference-pair logging or fine-tune pipeline. This is forward-looking roadmap, not built.

If OGZ_Simple represents the *intended* direction, the cross-compare/learning system and ALLaM are the largest unbuilt pieces of that vision. If the TechDoc remains the contract, these are out of Phase 1–3 scope and only the matrix/chains/fal.ai overlap matters.

6.2 Deliberate architecture decisions (documented in CHANGELOG)

- fal.ai** replaces **Weavy.ai** across the entire visual layer (image + video).
- CEO** → **OpenAI**, **COO** → **DeepSeek** (from Claude Sonnet/Haiku) due to Anthropic budget; CCO/Copilots unchanged.
- negative_pattern_library** → **global_negative_patterns**; **leads table dropped** (marketing form removed).
- These are real and reasonable, but they diverge from the signed TechDoc and should be ratified as ADRs so milestone acceptance is not contested.

7. Consolidated Gap Analysis — What Is Missing / Incomplete

Phase	Missing / Incomplete Item	Severity	Why it matters
1	Stripe payments & Upgrade Flow	HIGH	No revenue path; posts 9–20 cannot be unlocked; SEC-10 untestable
1	PostHog analytics instrumentation	MED	No funnel/upgrade-trigger data; listed in Phase-1 stack
1	Hard-coded secrets & ngrok URLs in n8n (SEC-02)	HIGH	Security non-compliance; tunnels break in prod
1	A02 on-demand flow still "Testing Mode"	MED	On-demand generation not on a production trigger
1	CEO/COO model vendor ≠ contract	MED	Needs client sign-off vs §13.1 acceptance
1	Weavy → fal.ai not ratified in contract	LOW	Documentation/acceptance alignment
1	Onboarding 15-question UX re-architected	LOW	Confirm matches client acceptance criteria
2	CIO (Gemini 2.5 Pro) pattern analysis	PHASE 2	Service is an empty placeholder
2	Meta Insights API	PHASE 2	No performance-data ingestion
2	Client Google Business OAuth	PHASE 2	Only public Places lookup exists

2	Postiz auto-publish end-to-end	PHASE 2	Connectivity built; scheduled publishing not wired
3	CARO Arabic QC (Jais/DeepSeek)	PHASE 3	Arabic QC still on GPT-based CCO
3	Advanced competitor analysis engine	PHASE 3	Basic capture only; no strategic analysis
3	(OGZ_Simple) Cross-compare + ALLaM + fine-tune loop	VISION	Largest unbuilt piece of the v2 narrative

8. Recommended Priorities

- 1. Close the Phase-1 launch blockers first:** implement Stripe Checkout + webhook verification (unblocks revenue & SEC-10), and remediate the hard-coded secrets / ngrok URLs in the n8n flows (SEC-02). Promote A02 to a production webhook.
- 2. Ratify the deviations:** add ADR entries (and obtain client sign-off) for fal.ai-vs-Weavy and the OpenAI/DeepSeek model swap, so M1/M2 acceptance is not contested under §13.
- 3. Add PostHog instrumentation** to complete the Phase-1 stack and enable upgrade-funnel tracking.
- 4. Confirm onboarding acceptance:** validate the multi-stage onboarding flow against the client's "15-question" expectation.
- 5. Phase 2 sequencing:** finish Postiz auto-publish, then build the CIO analyzer (Gemini) and Business-Profile OAuth / Meta Insights ingestion.
- 6. Phase 3:** video chains are done — focus on the CARO Arabic-QC swap and a real competitor-analysis engine. Decide explicitly whether the OGZ_Simple cross-compare/ALLaM learning loop is in scope.

OpenClaw Phase 1–3 Gap Analysis · Prepared 13 June 2026 · All status determinations are evidence-based and cite repository paths. Where the two requirement documents conflict, the OpenClaw_TechDoc_v1.0 (declared "single source of truth", §14.3) governs Phase definitions; OGZ_Simple is treated as evolved intent. Percentages are indicative completion estimates, not contractual measures.